MILITARY SPECIFICATION

OBJECTIVE LENS ASSEMBLY FOR AN/PAS-7

This specification is approved for use by USA Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the objective lens assembly for Viewer, Infrared AN/PAS-7 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall
PPP-C-843	- Cushioning Material, Cellulosic
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic, (and Connectors)
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116	- Preservation-Packaging Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	- Marking of Electronic Items

STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-774007	- Objective Lens Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. The objective lens assembly, one of the assemblies comprising Viewer, Infrared AN/PAS-7, consists of a three element optical system. The three elements, two silicon and one germanium are contained in a single assembly; held in relative position to each other by temperature compensating spacer; protected against fogging through internal venting and are coated with an anti-reflective coating.

3.2 First article. When specified the contractor shall furnish sample unit(s) for first article inspection and approval, (see 4.3 and 6.2).

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3.3 Construction. The objective lens assembly shall be constructed in accordance with the drawings, parts list and other documents listed on USAECOM Drawing DL-SM-B-774007.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified, shall be selected by the contractor, shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Operational and environmental.

3.5.1 Focal length. The objective lens assembly shall have an effective focal length of 2.5 inches, plus or minus 3 percent.

3.5.2 Resolution. The objective lens assembly shall be corrected so that for any point within a half field no greater than 3° at least 82 percent of the energy collected and transmitted shall form a circle of confusion whose diameter is no greater than 0.004 inch (1.6 milliradians). This resolution shall be achieved in a flat focal surface with the objective lens assembly focused for an object at infinity.

3.5.3 Transmittance. The objective lens assembly shall have an average transmittance of not less than 78 percent over the spectral region of 3 to 5 microns.

3.5.4 Temperature.

3.5.4.1 Operating. Minus 32°C to plus 49°C.

3.5.4.2 Non-operating. Minus 62°C to plus 68°C.

3.5.5 Temperature shock. Minus 57°C plus 0°C, minus 1.5°C to plus 68°C, plus 1.5°C, minus 0°C within five minutes.

3.5.6 Vibration. The objective lens assembly shall withstand vibratory displacement of 0.030 inch peak to peak minimum over a frequency range of 10 to 55 Hz.

3.6 Government furnished equipment. Unless otherwise specified in the contract or purchase order, there shall be no Government furnished equipment provided.

3.7 Treatment and painting. Unless otherwise specified, the metallic parts of the objective lens assembly shall be given a treatment and painting per MIL-STD-171.

3.8 Marking. Unless otherwise specified, the objective lens assembly shall be marked per MIL-M-13231.

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3.9 Workmanship. Unless otherwise specified workmanship of the objective lens assembly shall conform to Requirement 9, MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.7).

4.3 First article inspection.

4.3.1 Inprocess inspection prescribed by a specification or drawing. Each lot of components and subassemblies shall have all parameters examined for defects in conformance with the inspection plan in the applicable specification or on the drawing. Defects not listed as major shall be classified as minor.

4.3.2 Examination. Each assembled objective lens assembly shall be examined for defects in accordance with Table I. One or more defects shall be cause for rejection.

TABLE I. Examination.

Inspection	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.3, 3.4	MIL-STD-252
Treatment and painting not as specified	3.7	MIL-STD-171
Markings not as specified or illegible	3.8	MIL-M-13231
Workmanship not as specified	3.9	MIL-STD-454

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4.3.3 Tests. Upon successful completion of the inprocess inspections and assembled examination specified in 4.3 the first article objective lens assembly shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall constitute first article failure.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Focal length	3.5.1	4.6.1
Resolution	3.5.2	4.6.2
Transmittance	3.5.3	4.6.3
Temperature operating	3.5.4.1	4.6.4
Temperature non-operating	3.5.4.2	4.6.4
Temperature shock	3.5.5	4.6.5
Vibration	3.5.6	4.6.6

4.4 Quality conformance inspection.

4.4.1 Inprocess inspection. Inprocess inspection on the objective lens assembly shall be as specified in 4.3.1.

4.4.2 Examination. Each assembled objective lens assembly shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection of that objective lens assembly.

4.4.3 Tests.

4.4.3.1 Group A inspection. Following successful completion of the inprocess inspection and assembled examination of 4.4.1 and 4.4.2, each objective lens assembly shall be tested in accordance with Table III in the order listed. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Focal length	3.5.1	4.6.1
Resolution	3.5.2	4.6.2

4.4.3.2 Group B inspection. Group B inspection shall be conducted on objective lens assemblies selected from units which have passed the tests in 4.4.3.1. Sampling of lots shall be in accordance with MIL-STD-105, AQL

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2.5 percent defective, inspection level S-3 combined. Tests shall be conducted in accordance with Table IV, and shall be performed in the order listed.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Transmittance	3.5.3	4.6.3
Temperature operating	3.5.4.1	4.6.4
Temperature non-operating	3.5.4.2	4.6.4
Temperature shock	3.5.5	4.6.5
Vibration	3.5.6	4.6.6
Focal length	3.5.1	4.6.1
Resolution	3.5.2	4.6.2

4.4.3.3 Reinspection of Group B sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B inspections may be accepted on contract or order provided all damage is repaired and the units are resubjected to and pass Group A inspection.

4.5 Test methods. Tests required to determine compliance with the requirements of this specification shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified all tests shall be run at 23°C.

4.6 Operational and environmental.

4.6.1 Focal length. Using a collimated chopped infrared source, a nodal slide, test apertures and a test detector, the focal length of the objective lens assembly shall be determined. Failure to comply with the requirement of 3.5.1 shall constitute a failure of this test.

4.6.2 Resolution. Using a chopped infrared source collimated at an apparent range greater than 500 meters, suitable test apertures, and a test detector with a sapphire window, the amount of energy passed by an 0.004, plus or minus 0.0002 inch diameter aperture relative to a larger 0.040, plus or minus 0.005 inch diameter aperture shall be determined. Initially, the inspection shall then be repeated at no less than four additional off-axis points, in a flat focal surface or image plane, within a half-field no greater than 3°. Failure to comply with the requirement of 3.5.2 shall constitute a failure of this test.

4.6.3 Transmittance. Using a collimated chopped infrared source, an off-axis collimated mirror (masked to a 2.5, plus or minus 0.010 inch diameter circle) with known reflectivity over the spectral region of 3 to 5 microns, and an indium antimonide or lead selenide detector filtered in this spectral region, obtain a peak output from the detector. Replace the

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off-axis mirror with the objective lens assembly and obtain a peak output from the detector. The transmittance shall be calculated by determining the ratio of the second signal to the first after the reflectivity of the off-axis mirror is considered. Failure to comply with the requirement of 3.5.3 shall constitute a failure of this test.

4.6.4 Temperature.

4.6.4.1 High temperature. The objective lens assembly shall be subjected to test in accordance with Method 501, Procedure I of MIL-STD-810. Temperature for Step 2 shall be plus 68°C. Temperature for Step 4 shall be plus 49°. Failure to operate or pass inspection of MIL-STD-810 shall constitute failure of this test.

4.6.4.2 Low temperature. The objective lens assembly shall be subjected to test in accordance with Method 502, Procedure I of MIL-STD-810. The temperature for Step 2 shall be minus 62°C and shall be maintained until temperature stabilization of the test item is reached. Temperature for Step 4 shall be minus 32°C. Failure to operate or pass inspection of MIL-STD-810 shall constitute a failure of this test.

4.6.5 Temperature shock. The objective lens assembly shall be subjected to test in accordance with Method 503, Procedure I of MIL-STD-810. The maximum temperature, Steps 1 and 4 shall be plus 68°C. Failure to operate or pass inspection of MIL-STD-810 shall constitute failure of this test.

4.6.6 Vibration. The objective lens assembly shall be operated under standard conditions and a record made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking operational requirements of the objective lens assembly at the conclusion of the vibration test. The objective lens assembly shall be subjected to a simple harmonic motion having a vibratory displacement of 0.030 inch minimum peak to peak, with the frequency being changed in discrete steps of 1 Hz and maintained at least 10 seconds at each frequency over the frequency range of 10 to 55 hertz. Failure to operate shall constitute failure of this test.

4.7 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A, maximum military protection.

5.1.1.1 Cleaning. Objective lens assembly - Part of Viewer, Infrared AN/PAS-7 shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Objective lens assembly - Part of Viewer, Infrared AN/PAS-7 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Objective lens assembly - part of Viewer, Infrared AN/PAS-7. Each unit shall be individually packaged method III as follows: Cushion each item individually with material conforming to PPP-C-843, type II, 1/4 inch in thickness. Place the cushioned item within a barrier bag fabricated of material conforming to L-P-378, type II. Closure may be accomplished by staples, tape, adhesive or heat seal. Place the bag with contents within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B, minimum military protection.

5.1.2.1 Cleaning, drying, preservation application and unit packaging. Each objective lens assembly - Part of Viewer, Infrared AN/PAS-7 shall be cleaned, dried, preserved and unit packaged as specified in 5.1.1.

5.1.3 Commercial packaging. Commercial packaging shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A, maximum military protection. A quantity of objective lens assembly - Part of Viewer, Infrared AN/PAS-7, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

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5.2.1.1 Palletization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.2 Level B, minimum military protection. A quantity of objective lens assembly - Part of Viewer, Infrared AN/PAS-7, packaged as specified in 5.1 shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2.1 Palletization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of objective lens assembly-Part of Viewer, Infrared AN/PAS-7, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The objective lens assembly, a major component of Viewer, Infrared AN/PAS-7, shall focus and direct infrared energy from the scanning mirror to an infrared sensor.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Levels of preservation-packaging and packing required (see 5.1 and 5.2).
c. When first article tests are to be conducted and the number of first article units (see 3.2, 4.3 and Table II). Time frame required for submission of first article test reports and time frame required for approval of first article test reports by the contracting officer.

6.3 Definitions.

6.3.1 Room temperature. Room temperature is defined as +23°C, ± 3°C.

Custodian:
Army - EL

User activity:
Army - MU

Preparing activity:
Army - EL

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-O-49086(EL) OBJECTIVE LENS ASSEMBLY FOR AN/PAS-7

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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